510(k) PREMARKET NOTIFICATION

CardioFlow Vascular Stent System

CardioMed Technologies, Inc.

Submission Date: 10/29/2025
Device Classification: Class II
Product Code: NIQ

EXECUTIVE SUMMARY

This 510(k) premarket notification is submitted to demonstrate substantial equivalence of the CardioFlow Vascular Stent System to the legally marketed predicate device XIENCE Sierra (K181351).

Device Name: CardioFlow Vascular Stent System
Manufacturer: CardioMed Technologies, Inc.
Classification: Class II
Regulation: 21 CFR 870.3450
Product Code: NIQ
Predicate Device: XIENCE Sierra (K181351)

DEVICE DESCRIPTION

The CardioFlow Vascular Stent System consists of:

- A balloon-expandable cobalt-chromium (L605) stent
- Proprietary drug-eluting polymer coating with sirolimus
- Rapid-exchange balloon catheter delivery system
- Stent dimensions: 2.5-4.0mm diameter, 8-33mm length
- Strut thickness: 81 ;ÆÐ
- Drug load: 100 ;Ær 6—&öÆ–Ðus per cm^2

Key Features:

- Enhanced deliverability through tortuous anatomy
- Optimal radial strength (16.0 psi)
- Minimal recoil (<5%)
- Controlled drug release over 180 days
- MRI conditional (1.5T and 3T)

INDICATIONS FOR USE

The CardioFlow Vascular Stent System is indicated for improving coronary luminal diameter in patients with symptomatic ischemic heart disease due to discrete de novo or restenotic lesions in native coronary arteries with:

• Reference vessel diameters of 2.5 to 4.0 mm
• Lesion lengths "d 30 mm

SUBSTANTIAL EQUIVALENCE

Comparison to Predicate Device

Feature	CardioFlow (Subject)	XIENCE Sierra (K181351)
Indications	Coronary artery disease	Coronary artery disease
Material	L605 Co-Cr	L605 Co-Cr
Drug	Sirolimus	Everolimus
Strut Thickness	81 ;ÆÒ	ƒ ¼m
Diameters	2.5-4.0 mm	2.25-4.0 mm
Lengths	8-33 mm	8-38 mm
Delivery System	Rapid Exchange	Rapid Exchange

The CardioFlow system demonstrates substantial equivalence through:

- Same intended use and indications
- Similar technological characteristics
- Comparable safety and effectiveness profile
- Equivalent clinical performance

PERFORMANCE DATA

Biocompatibility Testing

All materials tested per ISO 10993 series:

- Cytotoxicity (ISO 10993-5): Passed
- Sensitization (ISO 10993-10): Passed
- Irritation (ISO 10993-10): Passed
- Acute Systemic Toxicity (ISO 10993-11): Passed
- Subchronic Toxicity (ISO 10993-11): Passed
- Genotoxicity (ISO 10993-3): Passed
- Implantation (ISO 10993-6): Passed
- Hemocompatibility (ISO 10993-4): Passed

Mechanical Testing

- Radial Strength: 16.0 ± 0.5 psi
- Recoil: 3.8 ± 1.0%
- Foreshortening: 1.2 ± 0.5%
- Flexibility: 0.45 N
- Fatigue Testing: 400 million cycles - No fractures

Clinical Performance

CardioFlow SPIRIT Clinical Trial Results:

- Primary Endpoint (12-month TLR): 3.2%
- Target Vessel Failure: 5.1%
- Cardiac Death: 0.8%
- Target Vessel MI: 2.1%
- Definite Stent Thrombosis: 0.4%

CONCLUSION

The CardioFlow Vascular Stent System has been shown to be substantially equivalent to the predicate device XIENCE Sierra (K181351) based on:

1. Same indications for use
2. Similar technological characteristics
3. Comparable materials and design
4. Equivalent performance testing results
5. Similar clinical outcomes

The minor differences in drug type (sirolimus vs everolimus) do not raise new questions of safety and effectiveness, as both drugs are well-established in the coronary stent market with proven safety profiles.

Therefore, the CardioFlow Vascular Stent System is substantially equivalent to the predicate device and suitable for clearance under 510(k).